

ROLE: you are a senior researcher in biophysics and experimental oncology with expertise in the interaction between non-ionizing, non-thermal electromagnetic fields and cancer cell biology. You are familiar with in vitro experimental design and the mechanistic literature on EMF–cell interactions.

OBJECTIVE: based exclusively on the evidence, patterns, and limitations described in the attached report (Electromagnetic fields as monotherapy on cancer cells in vitro: a literature-based dataset analysis), generate one well-grounded mechanistic hypothesis for the anti-tumoral action of non-thermal, non-ionizing EMFs as a mono-therapeutic modality, together with a corresponding experimental design proposal.

The hypothesis must be of a quality suitable to be submitted to domain experts (biophysicists, radiobiologists, experimental oncologists) for a critical review.

CONSTRAINTS:

- Read the report and, based on the findings, generate a hypothesis to test in a future real experiment. You need to explicitly justify your claims.
- Develop the hypothesis in depth rather than enumerating multiple superficial alternatives. Sub-mechanisms within the chosen hypothesis may be articulated as subsections only when they represent connected steps of the same causal chain, not as parallel alternative hypotheses.
- All claims must be traceable to evidence reported in the attached document. Do not import external data, studies, or numerical values that are not present in the report. Where the report identifies a gap or limitation, treat it as a constraint to be addressed by the experimental design rather than filled with assumptions.

STRUCTURE OF THE OUTPUT DOCUMENT:

1. Rationale and scope. Explain the choices made in this analysis and the reasoning behind them.
2. Mechanistic hypothesis. It is the proposed causal chain from field–cell interaction to a defined cell-fate outcome. Use subsections if the hypothesis articulates into distinct but connected mechanistic axes. Make explicit which steps are supported by direct evidence in the report and which are inferred links.
3. Parameter selection strategy. Justification of frequency, intensity, waveform, and exposure duration that you have decided to test. Indicate the biophysical or cell-type-specific rationale where applicable.
4. Experimental design.
 - Cell lines: which cancer lines and which healthy counterparts to test, with explicit justification.
 - Exposure apparatus: the most suitable setup for the chosen field type, with the controls needed to isolate the non-thermal component.
 - Outcomes to measure: a core set covering cell-fate outcomes, mechanistic outcomes, and molecular markers, aligned with the axes invoked in the hypothesis.
 - Controls and selectivity test: how the cancer-vs-healthy comparison will be carried out.
5. Predictions. The expected pattern of results if the hypothesis is correct (which combinations of parameters and cell contexts should produce anti-tumoral effects, which should not, and what would falsify the hypothesis).

OUTPUT FORMAT: deliver the result as a .docx file of maximum 5 pages. Be dense and precise; avoid filler, restatement, and generic background. Write in a tone appropriate for expert peer review.